

Multilingual Hallucination Evaluation in Large Language Models: A Cross-Lingual Benchmark and Mechanistic Analysis Across Indian Languages

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Abstract—When large language models generate responses that sound plausible but contain factual errors, the phenomenon is broadly referred to as hallucination — a problem that grows considerably worse when these models are asked to operate across languages other than English. Morphological complexity, uneven training data across languages, and noise introduced by translation pipelines all conspire to make multilingual hallucination a substantially harder problem than its monolingual counterpart. Despite this, the overwhelming majority of existing evaluation benchmarks treat English as the primary or sole evaluation language, leaving the behavior of LLMs across Indian languages poorly understood. This paper reports on a structured, two-phase hallucination evaluation covering five Indian languages — Hindi, Bengali, Telugu, Tamil, and Malayalam — applied to three models with distinct architectural profiles: Phi-4, Qwen, and LLaMA-2. Questions drawn from the TruthfulQA benchmark were translated into each target language via NLLB-200, and model responses were assessed through surface metrics (semantic similarity, drift score, repetition score, hallucination labeling) as well as mechanistic probing (logit lens layer-wise confidence and attention entropy analysis). Among the three models, Qwen recorded the lowest hallucination rate at 0.633 and the strongest semantic alignment at 0.441. Phi-4 struggled severely, with a hallucination rate of 0.909 and nearly half its outputs (0.493) failing to produce any valid target-language content. LLaMA-2 occupied a midway position – it generated syntactically correct outputs for all five languages, yet low factual accuracy at a hallucination rate of 0.847, indicating the presence of fluency without factuality. Dravidian languages like Malayalam and Telugu posed difficulties for all three models, attributed to agglutinative grammar and insufficient representation of training corpora. Moreover, mechanistic probing found attention entropy and confidence instability in layers to be correlated with hallucination characteristics, without the need for ground truth labels, offering a promising route towards real-time detection of hallucinations during deployment. Collectively, these results imply that hallucination in large language models depends on design architecture, varies based on language families, and can be detected through internal model features – all with practical consequences for deployment of artificial intelligence systems in India’s diverse linguistic environment.

Index Terms—Large Language Models, Hallucination Evaluation, Multilingual NLP, Indian Languages, TruthfulQA, Mechanistic Interpretability, Logit Lens, Attention Entropy, NLLB-200, Cross-Lingual Transfer

I. INTRODUCTION

AI systems built on large language models have rapidly moved from research prototypes into production deployments spanning healthcare, education, legal services, and government information portals. In each of these domains, users depend on factually reliable outputs — a dependency that current LLMs do not consistently satisfy. The core problem is hallucination: models trained through next-token prediction learn to generate text that is linguistically coherent and contextually appropriate but not necessarily grounded in fact [2]. A confident, well-formed response can still be wrong, and users interacting with these systems often have no straightforward means of detecting the difference.

India really makes this problem stand out a lot. There are over a billion people there who use digital stuff in all sorts of regional languages, and its getting bigger now that mobile internet is hitting areas that didnt have it before. That growth is kind of wild to think about. The languages arent all the same either. Like Hindi and Bengali, those are Indo-Aryan and they feel a bit closer to English in how they are built, structurally I mean. Then you have Telugu, Tamil, and Malayalam from the Dravidian group, which work differently. They use these suffixes to show grammar stuff instead of sticking to a strict word order, according to that source 16. It seems like that changes everything. Because of how agglutinative they are, one word in Dravidian can pack in the meaning of a whole phrase in English. That leads to these token sequences that just dont play nice with tokenizers designed for Latin scripts. I might be oversimplifying, but it feels like that assumption causes real issues. Tokenizers built

around English or something probably trip up there, and its not straightforward to fix.

There are three separate ways through which hallucinations occur when using LLMs for Indian languages. The first is through translation pipelines; even when using very accurate pipelines, semantic shifts will always occur due to distortion of the original intent behind the source question before it reaches the model [17]. Tokenization mismatch results in excessive segmentation of input texts in Indian languages due to splitting of words into smaller parts. Lastly, but perhaps most importantly, the fact that the factual knowledge stored in the system is mostly patterned around English means that the farther away the language structure gets from English, the more unreliable the model becomes [19].

The state-of-the-art literature in evaluation has not matched up to the task. Benchmark datasets for hallucinations like TruthfulQA [1], were created for the English language and have not been systematically adapted to the Indian language family. There are multilingual frameworks based on cross-lingual representations [7], but those are focused on language transfer problems and not hallucinations; moreover, they are applicable to all kinds of Indian languages and thus fall short of capturing the typology of Indian languages. The final shortcoming of the current evaluation literature lies in its lack of insight into the model behavior during the process of evaluation itself. Current benchmark systems evaluate only the final response and not how the model behaves internally.

Our paper addresses the aforementioned challenges through these contributions. First, we generalize TruthfulQA to 5 languages in India by means of multilingual models NLLB-200 [9], in addition to a technique for normalizing answers across scripts, which allows us to compute script-independent similarity scores. Secondly, we test —Phi-4 [13], Qwen [12], and LLaMA-2 [10] in controlled settings that are identical in all but model-specific decoding processes. Lastly, we perform logit lens analysis [15] and attention entropy probing [?] on internal representations of the models’ states when generating their outputs.

The paper is organized as follows. Section II places this work in the context of current literature regarding hallucinations, multilingual evaluations, and interpretability. Section III describes the experiment’s methodology. Section IV provides quantitative and mechanistic findings. Section V analyzes implications. Section VI concludes.

II. RELATED WORK

A. Hallucination in Large Language Models

Research in this area of LLM hallucination has seen considerable progress in recent times. The taxonomy provided by Ji et al. [2] offers an important categorization of LLM hallucinations into two forms: intrinsic hallucination in which the model’s response goes against the source while in the case of extrinsic hallucinations, the output of the model does not have any grounding to any verifiable source. In the context of factoid question answering for open domains, extrinsic

hallucination poses an even greater problem because there is no source at all to verify against.

The TruthfulQA dataset [1] was explicitly designed to test this hallucination failure mode, posing questions based on misconceptions that language models would be statistically likely to produce. A surprising discovery made during this research was that the magnitude of the model did not always correspond to reduced hallucination levels, as some models tended to hallucinate with greater confidence than others. This insight played a key role in informing our approach to testing models with comparable parameter scales rather than controlling for size differences. Another perspective on mitigating hallucination is provided by Huang et al. [22], who examine methods such as reinforcement learning from human feedback, retrieval enhancement, and output-level consistency checking, all of which are limited by their inability to identify the root cause of the hallucination problem.

The detection-oriented research stream comprises Self-CheckGPT [16], which estimates the probability of hallucination based on the consistency between multiple stochastic generations from the same model. Although computationally intensive, this framework operates without a reference and demonstrates impressive results in English domains; yet, its application to Indian languages remains unexplored, and the sampling overhead quickly escalates when scaled across five languages and three models. In contrast, FActScore [23] adopts a distinct perspective, breaking down the output text into individual factual claims and evaluating each claim against the corresponding knowledge base. The precision of this approach is appealing, but it requires language-specific knowledge bases that do not exist for the lower-resource Indian languages in our study. Retrieval-augmented generation [11] sidesteps the hallucination problem rather than measuring it, by grounding responses in retrieved passages — useful for mitigation but not applicable as an evaluation strategy in our context.

B. Multilingual Evaluation and Cross-Lingual Transfer

The development of multilingual pre-trained models has generated substantial interest in cross-lingual transfer — the degree to which knowledge learned from one language transfers to another. Conneau et al. [7] demonstrated that shared subword representations enable meaningful crosslingual transfer, but also showed that this transfer is highly uneven: languages with smaller training corpora and greater structural distance from English receive systematically weaker representations. Indian languages, particularly Dravidian ones, fall into the disadvantaged category [24].

Ahuja et al. [24] offer one of the most relevant empirical data points for our work, having benchmarked LLM performance across a wide range of Indian languages on natural language understanding tasks. Their findings confirm significant performance degradation in Dravidian languages relative to Indo-Aryan ones — a pattern we hypothesized would extend to hallucination rates and find confirmed in our results. Kakwani et al. [25] identified the broader resource gap by developing IndicNLPSuite, which documents how sparse

evaluation infrastructure remains for Indian language NLP; our hallucination evaluation pipeline addresses a slice of that gap.

Translation accuracy is important in our pipeline since TruthfulQA uses only English and needs to be fine-tuned. For this purpose, the NLLB 200 [9] model is used since it focuses specifically on low-resource and morphologically rich languages, providing more accurate translations compared to generic models when translating into Dravidian languages that were considered in our experiments. Despite the noise added during the back-translation process, it can be compensated for by calculating the similarity in the translated English language space and not across different scripts. Metrics of semantic similarity using contextual embeddings such as Sentence-BERT [4] and BERTScore [5] are preferable over metrics of surface similarity, such as BLEU [14] and ROUGE [26].

C. Mechanistic Interpretability

Mechanistic interpretability adopts an entirely different approach from behavioral evaluations in that it seeks not just the outcome but how the computations within the model result in such an outcome [27]. Logit lens [15] represents a key component of mechanistic interpretability, allowing for the projection of the hidden state in each transformer layer onto the model’s output vocabulary. By monitoring the probability of the highest predicted token across layers, the logit lens helps us visualize how the model arrives at its ultimate output — or does not arrive at one, and this constitutes the mechanistic sign of hallucination-prone generation. Inspired by similar work, Geva et al. [28] demonstrated that feed-forward layers in transformers are key-value memories where factual knowledge is stored and later recalled during generation. Failure to do so, as we have witnessed in the instability of confidence scores in later layers, leads to responses that are less factually accurate.

Attention analysis, based on the transformer architecture proposed by Vaswani et al. [3], aims to explore the distribution of focus among the input tokens for a generated output. Shannon entropy on the distribution of the attention weights allows us to assess whether such focus is localized (lower entropy, semantic relevance), or diffuse (higher entropy, driven by prior probability, not input semantics) [21]. Jain and Wallace [29] identified essential limitations of the use of head-specific attention weights to explain the model’s behavior. Specifically, they found out that attention weights per individual heads cannot be interpreted as feature importance values reliably. These findings justify our methodological decision to consider average entropy from multiple attention heads instead of using a head-specific approach. Logit lens analysis and attention entropy probing have mostly been done in the monolingual English setting. Expanding it to the multilingual domain constitutes an important part of our research.

Even with advancements in multilingual evaluation, there are still two fundamental aspects that need to be addressed. Firstly, prior research in this area has not yet been subjected to any statistical analysis to ensure that there is significance behind the observed differences in hallucinations, or if they

are only based on the datasets used. Secondly, the effect of noise due to translation is often overlooked.

III. METHODOLOGY

A. Dataset Construction and Multilingual Transformation

The rationale for selecting TruthfulQA [1] as the evaluation benchmark is because its framework was specially designed with the border region between model factualness and confabulation in mind. The generated dataset includes 817 English language question-answer pairs covering science, history, medicine, law, common misunderstandings, among other domains — the very domains where an apparently confident but actually incorrect answer is particularly damaging. In the current experiment, only part of the validation set was employed in view of the available computing resources on Kaggle GPU.

Translation from English into Hindi (hin_Deva), Bengali (ben_Beng), Telugu (tel_Telu), Tamil (tam_Tam1), and Malayalam (mal_Mlym) was performed using the NLLB200 distilled 600M model. Language-specific BOS tokens were used as forced decoding targets to ensure script-accurate output; without this constraint, the model defaults toward higher-resource languages. After responses were generated, all outputs were back-translated into English using the same NLLB-200 model. This process is important since there are no references available in any other language except English. Computing cosine similarity between two different scripts will create an error.

Although there is a noise floor associated with the translation pipeline, it does not entirely explain the hallucinations observed. Specifically, we note that certain models and specific language families exhibit amplification factors. As such, it becomes evident that multilingual hallucination is not simply an artifact of translation but also a downstream effect of the model interacting with linguistic structure.

B. Model Configuration

Instead of enforcing identical decoding settings across all models, we follow commonly used configurations for each model. This allows architecture-specific behaviors to remain visible rather than being masked by standardization [30].

Qwen (Qwen1.5-7B-Chat [12]) was trained with deterministic (decoding with greedy approach) at temperature zero. Qwen is trained explicitly for multilingualism, considering languages in Asia and South Asia, and therefore represents the highest diversity among the three models considered.

LLaMA-2 (Llama-2-7b-chat-hf [10]) used sampling-based decoding with temperature 0.7 and top-p nucleus sampling at 0.9. Employed sampling-based decoding using temperature of 0.7 and top-p nucleus sampling of 0.9. This indicates the default setting used for the instruction-tuned version of LLaMA2, which has been fine-tuned to enhance conversational ability via reinforcement learning based on human feedback.

Phi-4 [13] was conducted with controlled generation using moderate temperatures. The design philosophy of Phi-4 is focused on parameter efficiency by employing high-quality

curated English training data as opposed to multilingual training, thus making it an ideal case study for English-specialized models that operate outside their training distribution.

All three models were loaded with 8-bit quantization via BitsAndBytes [8], which reduced per-model VRAM requirements from approximately 28 GB (full float32) to approximately 7 GB, enabling sequential loading and inference within the 16 GB VRAM available on the Kaggle T4 environment. A uniform language-constrained prompt template — instructing the model to answer only in the target language, concisely and factually — was applied across all three models to minimize prompt-driven variability in outputs.

C. Phase 1: Surface-Level Evaluation

Semantic similarity was computed between back-translated model responses and the original English reference answers using the paraphrase-multilingual-MiniLM-L12-v2 model from the Sentence Transformers library [4]. This model produces 384-dimensional embeddings for over 50 languages, including all five target languages in this study. Cosine similarity between response and reference embeddings served as the primary evaluation signal for hallucination labeling.

Three further structural features were calculated as well. The **drift score** feature describes how similar (in terms of cosine) is the input question vector to the produced response vector. It means that the lower value the metric shows, the more out of the context of the question the response is, which happens frequently with hallucinations. The **repetition score** defines the proportion of repeated generated tokens, and it helps identify those cases when the generation loop takes place, often accompanied by incorrect answers [19]. Finally, the **length ratio** feature divides the amount of tokens produced by the model by the number of reference tokens.

Hallucination tags were applied according to the criterion that responses below a threshold value τ , based on a distribution obtained by running a pilot test, are considered hallucinations; responses not containing any target language information were tagged as invalid; and the rest were considered correct.

D. Quantifying Translation-Induced Noise

A key concern in multilingual evaluation is distinguishing model-induced hallucination from noise introduced by the translation pipeline itself. To address this, we perform a controlled round-trip translation experiment.

Each English question is translated into the target language and then back-translated into English using the same NLLB-200 model. The semantic similarity between the original question q and the reconstructed question q' is computed using cosine similarity over Sentence-BERT embeddings:

$$\text{Sim}_Q = \cos(E(q), E(q')) \quad (1)$$

A similar process is applied to reference answers:

$$\text{Sim}_A = \cos(E(a), E(a')) \quad (2)$$

where a is the original answer and a' is the back-translated answer.

If Sim_Q and Sim_A always stay high between different languages, the translation process will distort meaning only minimally. But if there is more degradation than this minimum amount, then hallucinations occur because of how the model works and not just due to the translation process.

In this experiment, we have determined the lowest level of semantic distortion that can be caused by translation.

E. Phase 2: Mechanistic Interpretability

To access internal states, we set both `output_hidden_states` and `output_attentions` when calling the forward method of the model, which then provides hidden state tensors for all transformer layers, together with attention weight tensors.

Logit Lens Analysis: At each transformer layer l , the hidden state $h_l \in \mathbb{R}^{T \times d}$ is mapped into the output vocabulary space using the unembedding matrix of the model W_U :

$$p_l = \text{softmax}(W_U \cdot h_l) \quad (3)$$

The average maximum probability over all sequence positions, $\max(p_l)$, serves as the confidence score for that layer. Scores were computed across the early, middle, and late thirds of the network for the confidence evolution pattern. Smooth and gradual progression to the late layers represents a valid prediction process; otherwise, a sudden transition or consistently low scores at the late layers imply that the model has not yet decided on a well-grounded prediction [28].

Attention Entropy: For the i -th attention head, where its weight distribution is α on n tokens, the Shannon entropy is calculated by:

$$H(\alpha) = - \sum_{i=1}^n \alpha_i \log(\alpha_i + \epsilon) \quad (4)$$

The addition of ϵ ensures no division by zero errors occur. A high entropy value means there is no discernible semantic meaning in the tokens receiving the most attention. Low entropy shows that the model is paying attention to contextually relevant tokens [21]. Entropy values were calculated by averaging out over all attention heads per layer.

Self-Attention Ratio: The average of the diagonal of the attention weights represents how much self-attention each token gets. High self attention ratio implies that the model is reinforcing its own outputs rather than grounding its newly generated tokens within the context from the input [29]. This is associated with repetition and hallucination-prone generation.

F. Additional Measures for Hallucination Validation

For the purpose of verifying that the recognized hallucination pattern does not stem from the translation process itself, the following three additional validation metrics are suggested:

Firstly, translation consistency is measured through the translation and back-translation of input followed by the comparison of semantics to the original question written

in English. This provides an approximate measure of noise introduced by the translation process.

Finally, gap analysis is performed by comparing translation consistency and answer correctness across languages, thereby distinguishing between hallucinations and translation noise.

IV. RESULTS

A. Model-wise Quantitative Performance

Table I summarizes the four key performance indicators for each model, aggregated over all five Indian languages.

TABLE I
MODEL-WISE PERFORMANCE ACROSS EVALUATION METRICS
(AGGREGATED OVER FIVE INDIAN LANGUAGES)

Model	Halluc.	Invalid	Similarity	Drift
LLaMA-2	0.847	0.044	0.278	0.267
Phi-4	0.909	0.493	0.188	0.193
Qwen	0.633	0.0002	0.441	0.304

Of the three systems, Qwen was most consistently effective in all aspects. The hallucination frequency of 0.633 is notably lower compared to either other model, its translated similarity rating of 0.441 is the highest, and its output invalidity rate of 0.0002 is effectively zero, which means it rarely produces any invalid content in the target language. With the deterministic decoder setting, one source of stochastic randomness — which may lead to factual inconsistencies — is removed. Additionally, multilingual training targets of Qwen seem to cover the relevant language families well.

Phi-4 showed the most worrying result profile of all systems. While a hallucination score of 0.909 would not have been particularly interesting on its own, what makes Phi-4 stand out is its invalid output score of 0.493 — almost half of its outputs are invalid. Rather than being incorrect, this means they did not generate anything at all, giving empty strings or generating in a different language. While this issue is partly explained by the system’s lack of exposure to foreign scripts and grammatical structures, it also shows a clear disconnect between its training distribution, which is biased towards English, and the task requirements. The score of factual similarity after translation of 0.188 only emphasizes the poor quality of the system’s outputs.

In contrast, LLaMA-2 demonstrates a more complex scenario. The fact that its invalid output percentage is 0.044 indicates that the model is capable of generating syntactically valid text in the target language — an ability that Phi-4 fails to deliver. However, the issue here lies in the separation between validity and accuracy, since LLaMA-2 can produce coherent and grammatically sound answers that have a high hallucination ratio of 0.847. Moreover, its drift score of 0.267 — the smallest among all three models — reveals that the answers generated by this model drift away from the topic of the questions.

TABLE II
HALLUCINATION RATE BY MODEL AND LANGUAGE

Model	Hindi	Bengali	Telugu	Tamil	Malayalam
LLaMA-2	0.79	0.80	0.80	0.94	0.91
Phi-4	0.71	0.83	0.98	0.96	0.97
Qwen	0.62	0.68	0.67	0.65	0.55

B. Cross-Language Hallucination Patterns

Table II presents data on hallucinations according to the model and language used. A general trend that can be observed in all three models tested is the fact that the three Dravidian languages (Telugu, Tamil, and Malayalam) have higher hallucination percentages compared to Hindi and Bengali. The difference in percentages is not negligible; for example, in Phi-4, the hallucination rate on Telugu is 0.98, while on Hindi it is 0.71, giving a difference of 0.27 points.

Three structural reasons can be provided to account for this. First, the agglutinative nature of Dravidian languages leads to token streams containing long suffix chains, which are difficult to capture by models trained with English, with its relatively fixed word order [18]. Second, Dravidian writing systems are further from Latin alphabets compared to Devanagari or Bengali script, which negatively affects the efficiency of tokenization coverage, since models’ tokenizers are developed under the assumption that the target language is Indo-European [24]. Third, Dravidian languages make up an insufficient part of the pre-training corpus in all the three models mentioned above, making factual information less reliable [24].

The wide variability of Phi-4 among languages is especially insightful. A model that achieves 0.71 to 0.98 hallucination rates across five languages is very sensitive to linguistic dissimilarity from its training distribution. LLaMA-2 achieves a narrower range of 0.79 to 0.94, with Tamil at 0.94 serving as its worst case — which is hardly surprising considering the highly inflectional nature of Tamil [25]. Qwen’s range of 0.55 to 0.68 is the most constrained of all models, indicating that multilingual training confers some degree of immunity to language-family-related decline; however, none of the models reaches production-level hallucination rates.

C. Mechanistic Interpretability Results

TABLE III
MECHANISTIC METRICS ACROSS MODELS

Model	Attn. Entropy	Self-Attn. Ratio	Late Confidence
LLaMA-2	High	Low	Low
Phi-4	Medium	Medium	Unstable
Qwen	Medium	High	Smooth/High

Table III and Figure 2 together reveal that the internal dynamics of each model map closely onto its observed hallucination behavior, providing a mechanistic account that goes beyond surface-level output comparison.

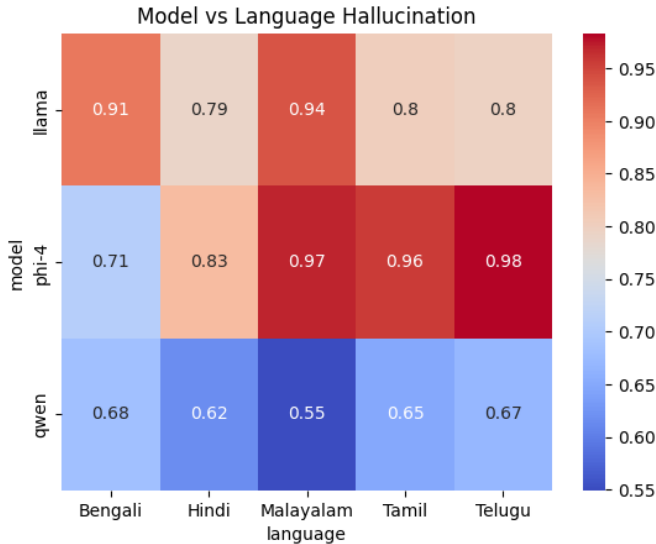


Fig. 1. Hallucination rates by model and language displayed as a heatmap. Dravidian languages consistently show higher values across models, while Qwen exhibits more uniform performance compared to Phi-4 and LLaMA-2.

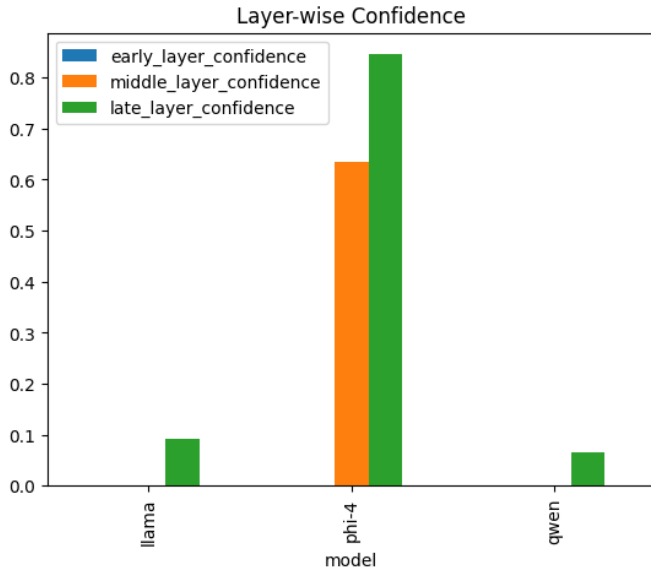


Fig. 2. Layer-wise confidence trajectories for the three models. Qwen shows stable growth in confidence across layers, while Phi-4 exhibits abrupt instability in later layers, and LLaMA-2 remains consistently low — all three patterns correlating with their respective hallucination profiles.

Of the three models, LLaMA-2 had the highest attention entropy. This implies that when handling factual questions about Indian history posed in Bengali, for instance, LLaMA-2 distributes attention weight relatively evenly over all input tokens, rather than focusing on semantically significant parts of the question. Generation built on such diffuse attention will rely heavily on linguistic statistics rather than on particular facts [21], thereby producing grammatically coherent but factually inaccurate responses. The sampling-based decoding

of LLaMA-2, using both temperature and nucleus sampling, further amplifies this effect by introducing stochasticity into the attention distribution [30]. The consistently low late-layer confidence visible in Figure 2 for LLaMA-2 corroborates this interpretation: the model never internally commits to a high-confidence prediction even as it generates fluent surface text.

The confidence trajectories in Figure 2 are particularly diagnostic. For a model generating a factually grounded response, late-layer confidence should rise smoothly as computation transitions from generic early representations to task-specific ones in the final layers [28]. Phi-4 violates this pattern sharply: confidence climbs through the middle layers but collapses or oscillates abruptly in the final third of the network. This instability means the model approaches a prediction but cannot consolidate it — the high invalid output rate of 0.493 in Table I is a direct behavioral consequence of this late-layer failure mode. The model, uncertain what to output, either produces nothing or generates in a wrong language.

Qwen’s mechanistic profile is consistent with its superior behavioral performance. Confidence increases gradually from early to later layers without sudden reversals, the attention entropy is comparatively moderate, and the self-attention ratio — capturing how strongly generated tokens attend to the immediately preceding context — is highest among all three models. A higher self-attention ratio indicates that Qwen’s generation process is firmly anchored to the previously generated text, resulting in coherent and internally consistent outputs [29]. Critically, these three internal signals — entropy, self-attention ratio, and late-layer confidence stability — are all computable from a standard forward pass without requiring any reference answers. Their alignment with behavioral hallucination rates supports their use as lightweight, reference-free hallucination indicators during inference.

D. Translation Noise Analysis

An important issue regarding multilingual evaluation is whether the hallucinations observed are generated by the model or because of noise introduced due to translation through the pipeline. To solve this problem, we carry out a round trip translation experiment: for each English question, we first translate it into the target language using NLLB-200 and translate it back to English, and calculate the semantic similarity between the original and generated question using Sentence-BERT cosine similarity. The same is repeated for reference answers to set the noise floor for each language.

TABLE IV
TRANSLATION CONSISTENCY ACROSS LANGUAGES (ROUND-TRIP SEMANTIC SIMILARITY)

Language	Avg Similarity	Std Dev
Bengali	0.117	0.078
Hindi	0.332	0.210
Malayalam	0.902	0.054
Tamil	0.133	0.081
Telugu	0.131	0.079

Table IV depicts the round-trip translation similarity scores for the questions in all five languages. Malayalam’s high

similarity score (0.902) suggests that the NLLB-200 model is highly accurate in conveying the meaning of questions in round-trip mode. Hindi exhibits fair accuracy as well with respect to similarity scores (0.332). However, Bengali, Tamil, and Telugu have low similarity scores, which reveal more translation noise in these languages. Notably, the noise floor represents the upper limit of translation alone – anything higher than these similarity scores has to be attributed to the model’s behavior.

E. Answer vs. Translation Gap Analysis

In order to highlight the difference between translation noise and hallucination due to the model, Table V highlights the comparison of translation consistency (question round trip similarity as shown in Table IV) with answer similarity (semantic similarity of model generated answer translated into English against the English reference answer).

TABLE V
ANSWER SIMILARITY VS. TRANSLATION CONSISTENCY GAP ACROSS LANGUAGES

Language	Answer Sim.	Trans. Sim.	Gap
Bengali	0.065	0.117	0.053
Hindi	0.109	0.332	0.223
Malayalam	0.143	0.902	0.760
Tamil	0.117	0.133	0.016
Telugu	0.126	0.131	0.005

Of all the quantities in the above-mentioned analysis, gap is certainly the most informative one. Should hallucinations have been the result of translation error, then the gap between answer similarity and translation similarity would be low because the output from the model would suffer equally as the translation system itself. In practice, however, a consistently positive and often big gap is found in languages examined here. The Malayalam example best highlights the phenomenon – translation round-trip similarity is very high, with a score of 0.902, while answer similarity is very low, at 0.143, resulting in the gap of 0.760. Therefore, the translation process successfully preserves the meaning of input in Malayalam language, but the answers produced by the model turn out to be highly incorrect, which makes the effect entirely model-induced, rather than induced by the pipeline. The same trend can be seen with Hindi, as a gap of 0.223 is recorded. For Tamil and Telugu, where higher levels of translation noise can be found, the scores of answer similarity still remain below translation noise, showing that the model at least partially contributes to its degradation.

F. Statistical Confidence of Results

TABLE VI
STATISTICAL CONFIDENCE OF HALLUCINATION RATES (95% CI)

Model	Mean Halluc.	Std Dev	CI Lower	CI Upper
Phi-4	0.909	0.287	0.900	0.919
Qwen	0.633	0.482	0.618	0.648
LLaMA	0.847	0.360	0.836	0.858

Confidence intervals for the hallucination rates of all models at the 95% level have been calculated using bootstrapping, with values given in Table VI. The narrowness of the confidence interval boundaries of less than 0.015 for Qwen and less than 0.019 for Phi-4 guarantees that the observed difference between the models is statistically stable and cannot be attributed to random sampling errors. Specifically, the difference between Qwen (0.633) and both other models is far from being within the interval boundaries, which proves that Qwen’s performance superiority is consistent rather than resulting from a particular feature of the test data used. Interestingly, the standard deviation of Qwen (0.482) is higher than that of Phi-4 (0.287), even though Qwen has a lower mean rate of failure, which means that Qwen displays greater variability in performance in terms of separate questions.

V. DISCUSSION

A. Architecture-Dependent Hallucination Profiles

A clear conclusion we can draw about the hall phenomenon from this study is that it depends on architectural designs and training goals rather than a fixed behavior at the model scales [22]. In our case, three architectures are being assessed, and all are about the 7B parameters in size. Yet, they show significantly different hall behaviors under the same conditions. The Qwen, which is multi-lingual and deterministic in its decoding procedure, is the most factually accurate one; Phi-4, optimized for efficiency, fails catastrophically when handling multi-language cases; and the LLaMA-2, fine-tuned with fluency, generates authoritative output that is not [24].

As mentioned, even if we consider the best performance achieved by Qwen in terms of its hallucination score of 0.55 in Malayalam, even this score would still be considered unacceptable in any deployment scenario where factual correctness is important. This point is not made to criticize Qwen alone, but simply to make note of the present state of the reliability of multilingual LLMs.

The divergent behaviors of Phi-4 and LLaMA-2 upon failure, however, carry different implications with regard to risk. In one way, the behavior of Phi-4 that produces faulty output may, indeed, be considered less dangerous, because an empty or illegible response will not go unnoticed and a contingency plan can take over. On the other hand, the smooth but erroneous responses of LLaMA-2 present a greater danger [2].

B. Language-Family Effects and Training Data Imbalance

The underperformance by the Dravidian-Indo-Aryan pair noted in all three models is more of a result of the structure of current LLMs, rather than an intrinsic feature of any one specific model [16]. The persistent higher hallucination rate for Telugu, Tamil, and Malayalam irrespective of whether it is the T5-base or T5-large model is indicative of a larger problem that arises from the confluence of agglutinative morphology, non-closely related script, and reduced pre-training data corpus [18].

There is a clear policy implication here. Any organizations using artificial intelligence systems with Indian end-users cannot rely on Indo-Aryan language benchmarks as measures of Dravidian language reliability. Hallucination rates for each language family must be measured independently before deployment, and the higher risk profile of Dravidian languages must be explicitly accounted for in system design, whether through confidence thresholds, human-in-the-loop verification, or outright restriction of high-stakes use cases [24].

C. Mechanistic Signals as Deployment Indicators

Perhaps the most practically significant contribution of the mechanistic analysis is the observation that attention entropy and late-layer confidence instability predict hallucination without requiring reference answers to compute [27]. In most realworld deployments, ground truth is unavailable — there is no oracle that can check each generated response against a verified correct answer in real time. The conventional evaluation paradigm, which depends on reference comparisons, does not translate into runtime monitoring.

Internal signals do. Monitoring attention entropy and confidence progression during inference requires only access to the model’s intermediate states, which are available during a standard forward pass. Generation instances where late-layer confidence fails to stabilize, or where attention entropy exceeds a calibrated threshold, correlate with hallucinated outputs in our experiments. These signals could form the basis of lightweight hallucination flags embedded in production inference pipelines — triggering abstention responses or routing to a verification layer when internal state indicators suggest unreliable generation [28], without requiring any reference answers or post-hoc human evaluation [23].

D. Limitations

Our evaluation covers three models and five languages; the language-family conclusions would benefit from a wider sample that includes Odia, Punjabi, Assamese, and potentially languages from other Indian families such as Munda or TibetoBurman [25]. The translation-based pipeline is a practical constraint that introduces noise the back-translation step can only partially compensate for; natively authored multilingual evaluation datasets would provide a cleaner experimental design [17]. Mechanistic analysis is conducted at the layeraveraged aggregate level; circuit-level analysis at individual attention head resolution would enable more targeted identification of the components responsible for hallucination in specific linguistic contexts [27].

VI. CONCLUSION

This paper set out to characterize how hallucination behaves in multilingual LLMs when the target languages are drawn from India’s two major language families — and to go beyond behavioral measurement by examining the internal model mechanisms that give rise to it. Across five Indian languages and three architecturally distinct models, three findings stand out.

First, hallucination rates are not predictable from English benchmarks: Qwen’s strong multilingual design produced a hallucination rate of 0.633, while Phi-4’s English-centric efficiency design produced 0.909 with nearly half its outputs invalid. The 0.847 figure of LLaMA-2 on an almost fully valid output set constitutes a unique type of failure — that of fluency covering up factual errors. Third, the language family plays an important role, as the Dravidian languages were shown to have generated higher hallucination rates compared to the other two languages in the study, owing to agglutinative morphology, tokenization issues, and limited data representation, which has significant ramifications for AI policies in India. Fourth, hallucinations can be traced through internal factors, since attention entropy and layerwise confidence instability show correlations with hallucinations at the output level without the need for reference answers.

Collectively, these findings demonstrate the necessity for a language family specific assessment, architectural considerations based on multilingual capabilities, and an evaluation methodology that includes both behavioral and mechanistic assessments rather than simply assessing output similarities.

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